

Coordinate Scaling for Raw PenroseDebruim Patches

Abstract

This note derives the exact part of the coordinate second moment that grows with K in the raw De Bruijn construction. The result is

$$V_{\text{index}}(K) = \frac{25}{6} K(K+1)$$

per Cartesian coordinate. The population analysis assumes $\mathbb{E}[x] = \mathbb{E}[y] = 0$ by symmetry and therefore identifies variance with raw second moment; neither patch means nor finite-sample means are subtracted. A seeded 100,000-condition sweep for every $K = 0, \dots, 5$ calibrates the bounded residual to the simple common value 5.65, giving

$$s_{xy}(K) = \sqrt{\frac{25}{6} K(K+1) + 5.65}.$$

The coefficient 25/6 is exact; 5.65 is empirical. A sixth uniform input rotates all five grid directions through one half of their angular spacing. This preserves the x/y second moment while making the wrapped output angle uniform on $[-\pi, \pi)$.

1 Raw construction

Write

$$\gamma = (\gamma_0, \dots, \gamma_4) \in [0, 1)^5$$

for the five phase values, and independently draw

$$\theta_{\text{offset}} \sim \text{Uniform}[0, 1].$$

Let K be a nonnegative integer and define the mapped angular offset

$$\theta_0 = \left(\theta_{\text{offset}} - \frac{1}{2} \right) \frac{\pi}{5} \in \left[-\frac{\pi}{10}, \frac{\pi}{10} \right].$$

The five unit normals are

$$u_j = \begin{pmatrix} \cos \theta_j \\ \sin \theta_j \end{pmatrix}, \quad \theta_j = \frac{2\pi j}{5} + \theta_0, \quad j = 0, \dots, 4.$$

For phase vector γ , family j contains lines

$$u_j^T x = n - \gamma_j, \quad k \in \{-K, \dots, K\}.$$

For every unordered pair $i < j$, and every ordered pair of line indices (k_i, k_j) , one rhombus is emitted. Hence

$$N(K) = \binom{5}{2} (2K+1)^2 = 10(2K+1)^2.$$

This includes $N(0) = 10$.

Let $x = x_{ij}$ be the intersection of the selected lines. The five region coordinates are

$$r_i = k_i, \quad r_j = k_j, \quad r_m = \lceil u_m^T x + \gamma_m \rceil \quad (m \neq i, j).$$

The raw rhombus center is

$$c_{ij} = \sum_{m=0}^4 r_m u_m + \frac{u_i + u_j}{2}. \quad (1)$$

No patch mean is subtracted from (1). In particular, a phase-dependent translation of the complete patch is part of the target.

2 Exact line-index variance

Let $\mathbf{k}$ be uniform on the $2K + 1$ integers $\{-K, \dots, K\}$. Symmetry gives $\mathbb{E}\mathbf{k} = 0$, and

$$\text{Var}(\mathbf{k}) = \frac{1}{2K + 1} \sum_{k=-K}^K k^2 = \frac{K(K + 1)}{3}.$$

This identity is exact, including at $K = 0$.

For a fixed family pair define

$$A_{ij} = \begin{pmatrix} u_i^T \\ u_j^T \end{pmatrix}, \quad q_{ij} = \begin{pmatrix} k_i - \gamma_i \\ k_j - \gamma_j \end{pmatrix}.$$

Then

$$x_{ij} = A_{ij}^{-1} q_{ij}.$$

Holding γ fixed and varying the independently enumerated indices, the index-induced covariance is

$$\text{Cov}_{\text{indices}}(x_{ij}) = \sigma_{\mathbf{k}}^2 A_{ij}^{-1} A_{ij}^{-T}, \quad \sigma_{\mathbf{k}}^2 = \frac{K(K + 1)}{3}.$$

If $\Delta_{ij} = \theta_j - \theta_i$, then

$$A_{ij} A_{ij}^T = \begin{pmatrix} 1 & \cos \Delta_{ij} \\ \cos \Delta_{ij} & 1 \end{pmatrix}$$

and therefore

$$\text{tr} \left(A_{ij}^{-1} A_{ij}^{-T} \right) = 2 \csc^2 \Delta_{ij}.$$

After using rotational balance to divide the trace equally between x and y , the shared per-coordinate intersection contribution is $\sigma_{\mathbf{k}}^2 \csc^2 \Delta_{ij}$.

3 Average over the ten family pairs

Five unordered pairs have separation 72° modulo 180° , and five have separation 144° , whose squared sine equals that of 36° . Consequently

$$\begin{aligned} \frac{1}{10} \sum_{i < j} \csc^2 \Delta_{ij} &= \frac{1}{2} (\csc^2 72^\circ + \csc^2 36^\circ) \\ &= \frac{1}{2} \left(\frac{8}{5 + \sqrt{5}} + \frac{8}{5 - \sqrt{5}} \right) = 2, \end{aligned}$$

where

$$\sin^2 72^\circ = \frac{5 + \sqrt{5}}{8}, \quad \sin^2 36^\circ = \frac{5 - \sqrt{5}}{8}.$$

Thus the pair-averaged intersection variance contributed by the line indices is exactly

$$2\sigma_{\mathbf{k}}^2 = \frac{2}{3}K(K+1)$$

per coordinate.

4 The dual-projection factor

The five normals form a unit-norm tight frame:

$$\sum_{m=0}^4 u_m u_m^T = \frac{5}{2} I_2. \quad (2)$$

This follows by expanding the matrix entries and observing that the sums of $\cos 2\theta_m$ and $\sin 2\theta_m$ vanish. Hence

$$\sum_{m=0}^4 (u_m^T x) u_m = \frac{5}{2} x.$$

The unquantized dual projection therefore multiplies distances by $5/2$ and coordinate variances by $25/4$. Its exact line-index contribution is

$$V_{\text{index}}(K) = \frac{25}{4} \frac{2}{3} K(K+1) = \boxed{\frac{25}{6} K(K+1)}. \quad (3)$$

5 Exact decomposition of the bounded terms

For each non-intersecting family define the ceiling remainder

$$\delta_m = \lceil u_m^T x + \gamma_m \rceil - (u_m^T x + \gamma_m), \quad 0 \leq \delta_m < 1$$

up to the immaterial endpoint convention. For the intersecting families, the line equations give

$$r_i = u_i^T x + \gamma_i, \quad r_j = u_j^T x + \gamma_j.$$

Substituting these identities into (1) and applying (2) yields the exact decomposition

$$c_{ij} = \frac{5}{2} x_{ij} + \underbrace{\sum_{m=0}^4 \gamma_m u_m}_{G(\gamma)} + \underbrace{\sum_{m \neq i,j} \delta_m u_m}_{Q_{ij}} + \frac{u_i + u_j}{2}. \quad (4)$$

The first term contains the exact growing contribution (3). The remaining terms are bounded for fixed phases, but bounded does not mean variance-free or independent:

- $G(\gamma)$ translates the raw patch as the phases change;
- each δ_m is a sawtooth function of the intersection and is correlated with x_{ij} ;
- the set of omitted indices in Q_{ij} depends on the family pair;
- the half-diagonal $(u_i + u_j)/2$ is pair-dependent; and
- cross-covariances among all terms in (4) contribute to the total coordinate variance.

6 Population moments and finite-sample estimators

Let $C = (X, Y)$ be the center obtained by first drawing $\Gamma \sim \text{Uniform}([0, 1]^5)$ and then selecting one tile uniformly from its complete raw patch. The population analysis adopts the fivefold-symmetry identities

$$\mathbb{E}[X] = \mathbb{E}[Y] = 0.$$

It follows that

$$\text{Var}(X) = \mathbb{E}[X^2], \quad \text{Var}(Y) = \mathbb{E}[Y^2],$$

and the shared population scale is defined from the pooled raw second moment:

$$s_{xy}^2(K) = \frac{\mathbb{E}[X^2] + \mathbb{E}[Y^2]}{2}. \quad (5)$$

Neither a generated patch mean nor a finite Monte Carlo sample mean is subtracted in (5).

For auditability, the sweep also records centered empirical quantities. With M observed tiles,

$$\bar{x} = \frac{1}{M} \sum_{m=1}^M x_m, \quad v_x = \frac{1}{M} \sum_{m=1}^M (x_m - \bar{x})^2, \quad q_x = \frac{1}{M} \sum_{m=1}^M x_m^2.$$

The columns `variance_x` and `sample_variance_x` use denominators M and $M - 1$, respectively; the same definitions apply to y and wrapped angle. The pooled xy sample is the concatenation $(x_1, \dots, x_M, y_1, \dots, y_M)$. Therefore

$$\overline{xy} = \frac{\sum_m x_m + \sum_m y_m}{2M}, \quad q_{xy} = \frac{\sum_m x_m^2 + \sum_m y_m^2}{2M}.$$

Its centered population and sample variances use denominators $2M$ and $2M - 1$. The pooled mean is retained as a diagnostic even though separate $\bar{x}$ and $\bar{y}$ are more geometrically informative.

The observed x - and y -means are at most about 7.0×10^{-4} in absolute value. Under the zero-population-mean symmetry assumption these are finite-sample estimation errors. They are reported but are not used to center q_x , q_y , or q_{xy} .

7 Raw second-moment sweep and residual fit

The reproducible sweep uses seed 20260823, 100,000 independent independent $(\gamma, \theta_{\text{offset}})$ draws for each $K = 0, \dots, 5$, and every tile in every raw patch. Thus the tile observation count is

$$M = 100,000 \cdot 10(2K + 1)^2.$$

The full statistics are stored in `tests/scaling/raw_statistics.csv`. The entries relevant to the coordinate scale are:

K	q_{xy}	exact leading term	residual
0	5.653308993	0	5.653308993
1	13.980624932	8.333333333	5.647291599
2	30.645533604	25	5.645533604
3	55.646613015	50	5.646613015
4	88.978783742	83.333333333	5.645450409
5	130.646957465	125	5.646957465

Here “residual” means

$$R_K^{\text{emp}} = q_{xy} - \frac{25}{6}K(K+1).$$

The exact analysis does not prove that this residual is constant. Empirically, however, all six values lie close to 5.65. The simplest calibrated model over the requested domain is a constant residual. Its least-squares estimate in variance space is the arithmetic mean 5.647525847; rounding to three significant digits gives

$$R_{\text{cal}} = 5.65.$$

This is deliberately not presented with spurious extra digits. Combining it with (3) gives the operational formula

$$\boxed{s_{xy}(K) = \sqrt{\frac{25}{6}K(K+1) + 5.65}}. \quad (6)$$

Its largest relative discrepancy from the six measured second-moment scales is approximately 0.03%, at $K = 0$; for $K = 1, \dots, 5$ it is below 0.01%. Runtime code evaluates (6) directly and never reads the statistics CSV.

8 Wrapped angle statistics

The CSV angle columns refer to the raw generated angle wrapped into $[-\pi, \pi)$, in radians, before the training multiplier $\sqrt{3}/\pi$ is applied.

Under the requested equal-count assumption, the construction produces ten equally weighted directions separated by $\pi/5$. The new independent variable rotates all of those directions by

$$\theta_0 \sim \text{Uniform}\left[-\frac{\pi}{10}, \frac{\pi}{10}\right].$$

The intervals swept by adjacent directions meet exactly at their endpoints. Their equally weighted union is therefore the uniform distribution on the whole wrapped circle:

$$\Theta \sim \text{Uniform}[-\pi, \pi).$$

Consequently,

$$\mathbb{E}[\Theta] = 0, \quad \text{Var}(\Theta) = \mathbb{E}[\Theta^2] = \frac{\pi^2}{3}.$$

After training scaling, the second moment is multiplied by

$$\left(\frac{\sqrt{3}}{\pi}\right)^2 = \frac{3}{\pi^2}.$$

Thus scaled angle has exact unit second moment in the population. The committed angle-histogram CSV divides $[-\pi, \pi)$ into twenty intervals of width $\pi/10$; each interval has population probability $1/20$. Finite-sample counts and means fluctuate around 5% and zero, respectively.

The global rotation also preserves $\|c_{ij}\|^2 = x^2 + y^2$ for every tile. Therefore it does not change the pooled xy second moment or the scale derived in the preceding sections.

9 Status summary

- $N(K) = 10(2K + 1)^2$ is exact.
- The generator emits raw centers and never subtracts a patch mean.
- Population x/y scaling uses raw second moments under $\mathbb{E}[X] = \mathbb{E}[Y] = 0$.
- The growing term $(25/6)K(K + 1)$ is exact.
- The common residual 5.65 is calibrated from the six seeded raw sweeps and is not an analytic identity.
- Runtime scale is (6); the CSV is statistics only.
- The sixth uniform input rotates all directions by $[-\pi/10, \pi/10]$, making wrapped angle uniform with zero mean and second moment $\pi^2/3$.